## Track B – Software-Based AI System
### Milestone 2 – Working ML Core

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| **Course:** CSC-361 Machine Learning | **Semester:** Spring 2025 |
| **Track:** Track B – Software-Based AI System | **Due Date:** 30 April 2025 |
| **Instructor:** Dr. Shafiq Ur Rehman Khan | **Max Marks:** 7 / 20 |
| **Mapped CLO:** CLO-2 | |

### Mapped Course Learning Outcome (CLO)

• CLO-2: Apply regression, classification, and clustering algorithms to solve standard machine learning problems.

### Milestone Objective

Students will implement and demonstrate a working ML model at the core of their system. This milestone evaluates the correct application of regression, classification, or clustering algorithms within a functional software system.

### Required Deliverables

- Written Report (PDF, 3–4 pages) with screenshots
- Presentation / Demo (7–10 minutes)
- GitHub repository with model training and application code

### Submission Requirements

- Working ML model (regression/classification/clustering) implemented and tested
- Data preprocessing pipeline (null handling, encoding, scaling)
- Training methodology with train/test split documented
- Evaluation using multiple appropriate metrics
- Basic system integration evidence (screenshots, demo, or API call)

## Evaluation Rubric — Total Marks: 7 / 20

| Criterion | Excellent (Full Marks) | Good (75%) | Satisfactory (50%) | Needs Work (25%) | Marks | CLO |
|---|---|---|---|---|---|---|
| ML Model Implementation | Correct, working regression/classification/clustering model with proper code structure | Model works with minor issues | Model partially implemented | No working model | 2.0 | CLO-2 |

**Semester: Spring 2025** **Max Marks: 7**
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**Group Name / No.:** **Submission Date:**

| Criterion | Excellent (Full Marks) | Good (75%) | Satisfactory (50%) | Needs Work (25%) | Marks | CLO |
|---|---|---|---|---|---|---|
| Data Preprocessing Pipeline | Full pipeline: null handling, categorical encoding, feature scaling all implemented | Most preprocessing steps present | Partial preprocessing only | No preprocessing done | 1.5 | CLO-2 |
| Evaluation Metrics | Multiple suitable metrics used; results clearly reported with interpretation | At least two metrics used; partial interpretation | Only one metric (e.g., accuracy alone) | No evaluation metrics used | 1.5 | CLO-2 |
| System Integration | Clear evidence of ML model integrated into application layer (UI, API, or backend) | Partial integration demonstrated | Integration planned but not demonstrated | No integration attempted | 1.0 | CLO-2 |
| Code Quality & GitHub | Clean, commented code; GitHub repo with proper structure and README | Code readable; README present but brief | Code functional but unorganized | Code absent or non-functional | 0.75 | CLO-2 |
| Demo / Presentation | Clear live demo; model behavior and metrics explained confidently | Demo works; partial explanation | Demo attempted; significant gaps | No demo or failed demo | 0.25 | CLO-2 |

## Marks Distribution Summary

| Criterion | Max Marks | CLO |
|---|---|---|
| ML Model Implementation | 2.0 | CLO-2 |
| Data Preprocessing Pipeline | 1.5 | CLO-2 |
| Evaluation Metrics | 1.5 | CLO-2 |
| System Integration | 1.0 | CLO-2 |
| Code Quality & GitHub | 0.75 | CLO-2 |
| Demo / Presentation | 0.25 | CLO-2 |
| Total | 7.00 | |

**Semester: Spring 2025** **Max Marks: 7**

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**Group Name / No.:** **Submission Date:**

**Important Notes:** All group members must be able to explain their individual contribution during viva. Plagiarism in report or code will result in zero marks for the entire group. Accuracy alone is not an acceptable evaluation metric — use multiple metrics appropriate to the ML task.